Using Data Analysis to Address Important Social Questions

Geometric Mean

geometric mean (also known as the **mean proportional**) is a mean or average which indicates a central tendency of a finite collection of positive real numbers by using the product of their values (as opposed to the arithmetic mean, which uses their sum).

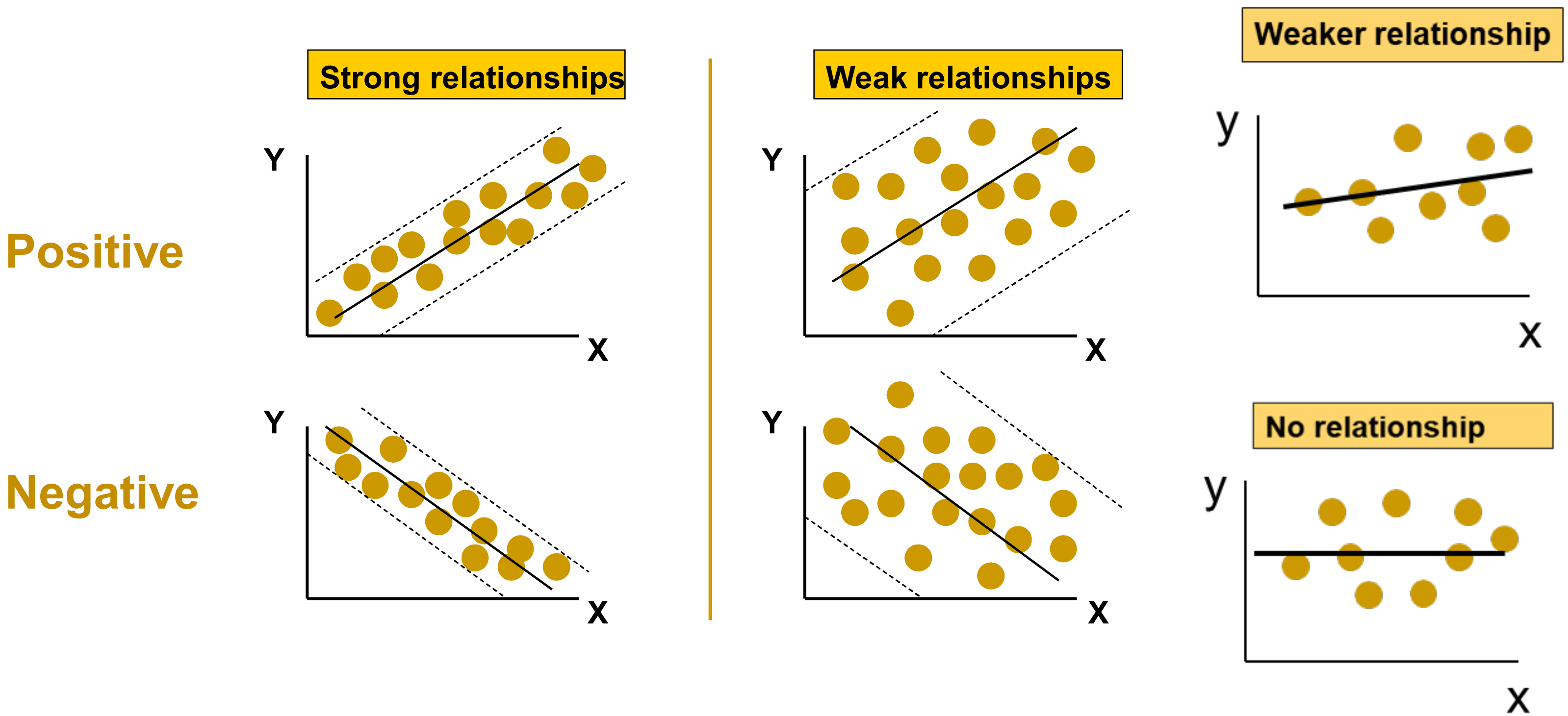
Geometric Mean: $\sqrt[3]{(3\times9\times27)} = \sqrt[3]{(729)} = 9$ *Arithmetic Mean:* $(3+9+27)/3 = 21$

Statistical measure used to calculate the central tendency used to find the central value of the data set in statistics especially when dealing with **growth rates or percentages**. This method is great for comparing **values that change over time**, like investment returns or population growth.

Relationships

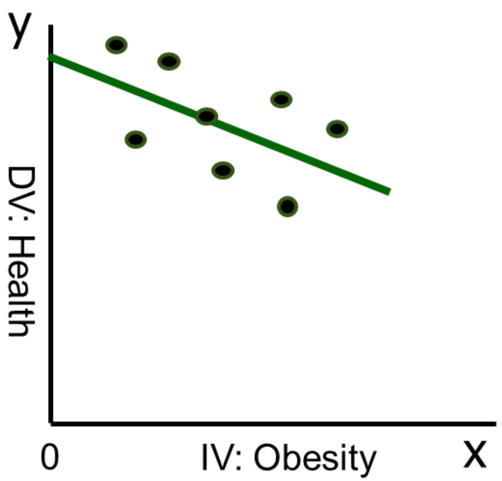
Relationship Statistic: Correlation

Consider a relationship using a scatterplot; a relationship between a dependent variable (Y) and an independent variable (X)



A relationship is NOT considered to be causal; **WHY?**

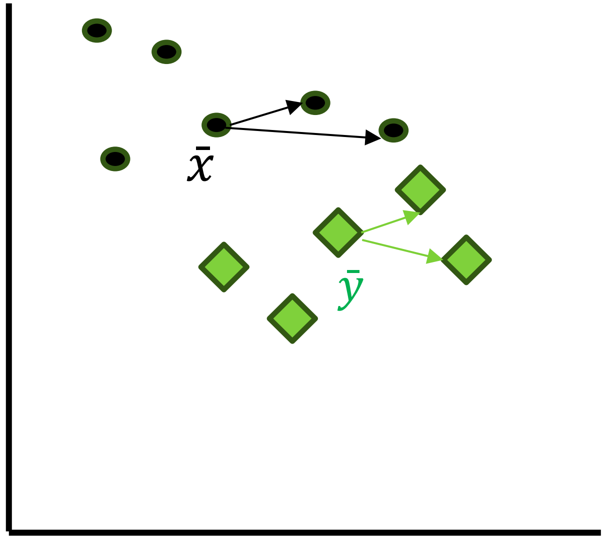
Recall our visualised Obesity – Health relationship



It might be useful to have a statistics that informs us about the nature of a distribution:

- **Dispersion of the variables about their averages:** Standard Deviation & Variance;
- **How a pair of variables, x & y, move together: Covariance;**
- **The relationship between two variables, x & y: Correlation.**

Relationship Statistic: Covariance

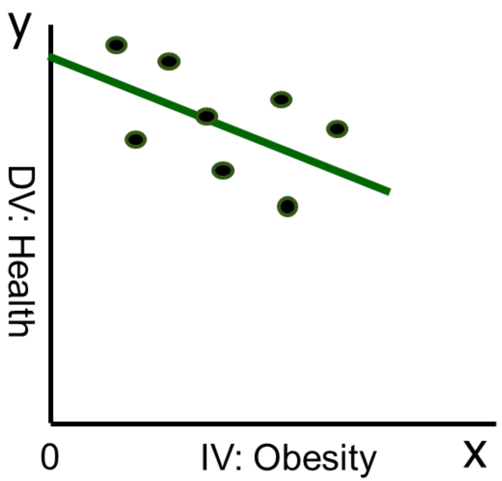


$$Cov(X,Y) = \frac{\sum(x-\bar{x})(y-\bar{y})}{n-1}$$

$$\begin{array}{cc} & \begin{array}{ccc} \mathbf{X_1} & \mathbf{X_2} & \mathbf{X_3} \end{array} \\ \begin{array}{c} \mathbf{X_1} \\ \mathbf{X_2} \\ \mathbf{X_3} \end{array} & \begin{bmatrix} Var(X_1) & Cov_{12} & Cov_{13} \\ Cov_{21} & Var(X_2) & Cov_{23} \\ Cov_{31} & Cov_{32} & Var(X_3) \end{bmatrix} \end{array}$$

Excel

Recall our visualised Obesity – Health relationship



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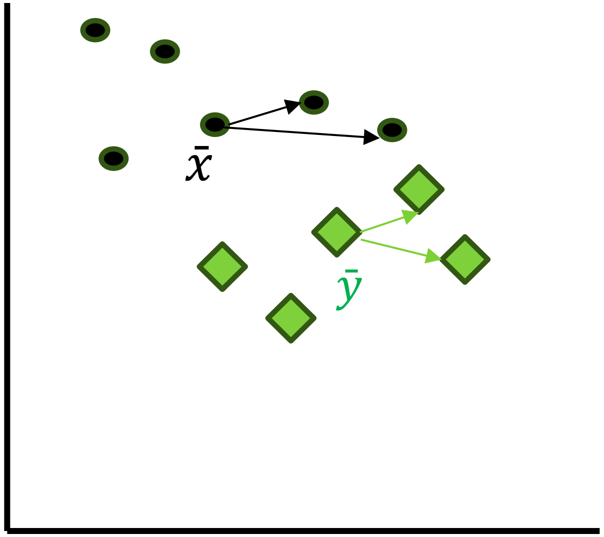
Correlation

This time we consider how the two variables move together, relative to one another:

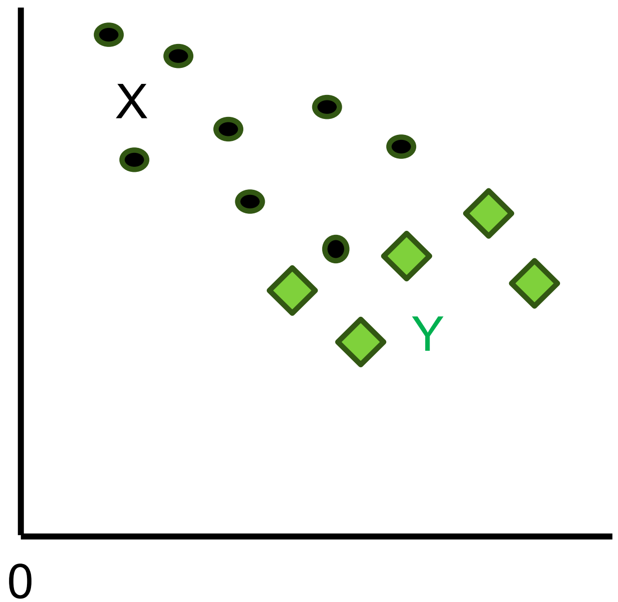


$$\frac{Cov(X,Y) = \dfrac{\sum (x-\bar{x})(y-\bar{y})}{n-1}}{S_x = \sqrt{\dfrac{\sum_{i=1}^{n}(X_i-\bar{X})^2}{n-1}} \; * \; S_y = \sqrt{\dfrac{\sum_{i=1}^{n}(Y_i-\bar{Y})^2}{n-1}}}$$

Excel

Relationship Statistics: Correlation

Graphically represent the relationship between x, y;



$$r_{(x,y)} = \frac{Cov(x,y)}{S_x * S_y}\text{ ; } (r = -1 \ldots 0 \ldots +1)$$

Calculate the Correlation of x, y

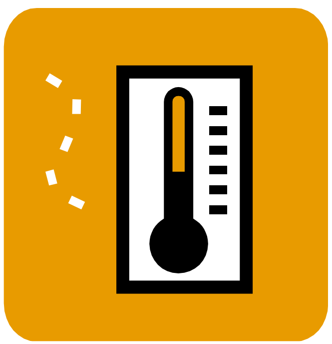
Interpreting Correlation Coefficient

r	Interpretation
r = -1	PERFECT negative linear relationship
-1 < r ≤ -0.7	STRONG negative linear relationship
-0.7 < r ≤ -0.3	MODERATE negative linear relationship
-0.3 < r < 0	WEAK negative linear relationship
r = 0	No relationship
0 < r < 0.3	WEAK positive linear relationship
0.3 ≤ r < 0.7	MODERATE positive linear relationship
0.7 ≤ r < 1	STRONG positive linear relationship
1	PERFECT positive linear relationship

Correlation (example)

- Example: Soft Drink Sales and Temperature

analysis of two numeric variables



Temp (°C)	Soft Drinks
14	60
22	40
28	100
30	100
36	75

Temp (°C) - X
Soft drink sale ($’000) - Y

Method: First calculate Covariance

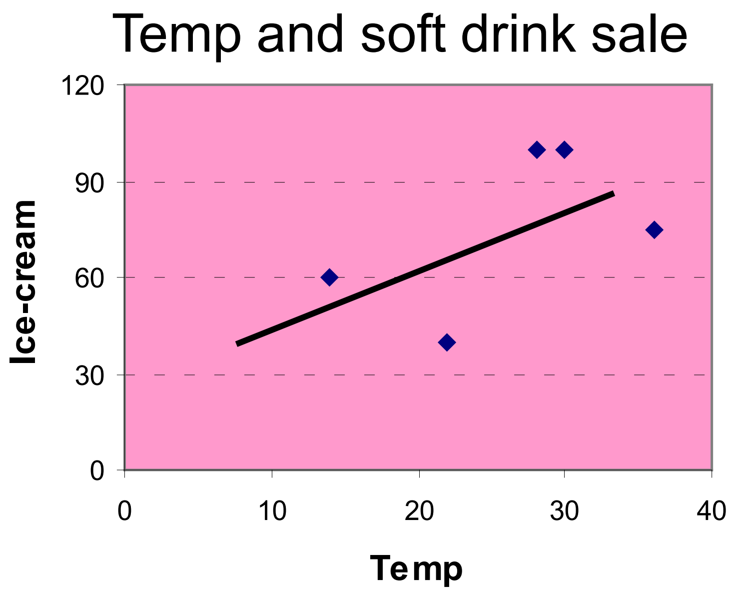
Covariance (Soft Drink Sales and Temperature)

Mean temp ($\bar{x}$)=26; mean soft drink sales ($\bar{y}$) = 75

X	Y	$(X-\bar{X})$	$(Y-\bar{Y})$	$(X-\bar{X})(Y-\bar{Y})$
14	60	**- 12**	**- 15**	**180**
22	40	**- 4**	**- 35**	**140**
28	100	**2**	**25**	**50**
30	100	**4**	**25**	**100**
36	75	**10**	**0**	**0**

$$Cov(X,Y) = \frac{\sum(x-\bar{x})(y-\bar{y})}{n-1} = \frac{470}{5-1} = 117.5$$

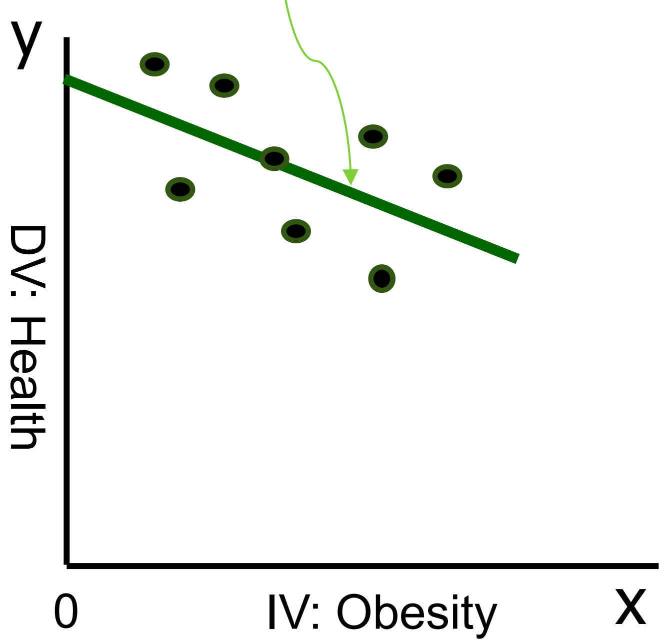
Interpretation of Covariance



<u>Covariance only informs us of the direction of association</u>

Method: Then calculate Correlation

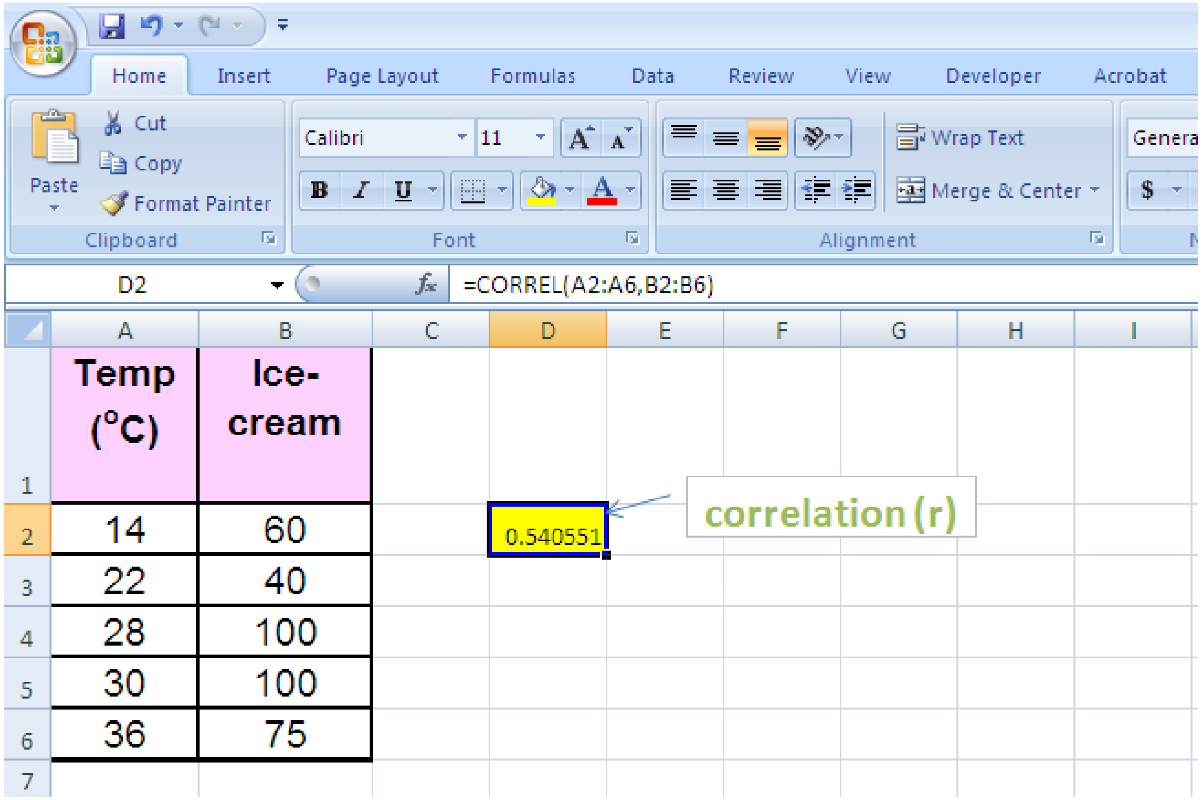
$$\text{Correlation}(X, Y) = r_{xy} = \frac{Cov_{xy}}{S_x S_y}$$



$$\text{Correlation}(Health, Obesity) = r_{health,obesity} = \frac{Cov_{health,obesity}}{S_{health} * S_{obesity}}$$

Task: Calculate the correlation between health and obesity and interpret the result

Excel: Covariance & Correlation



Task: Graphically represent the relationship between Obesity (x) & Health (y);
Calculate the Correlation of (x, y)

Excel

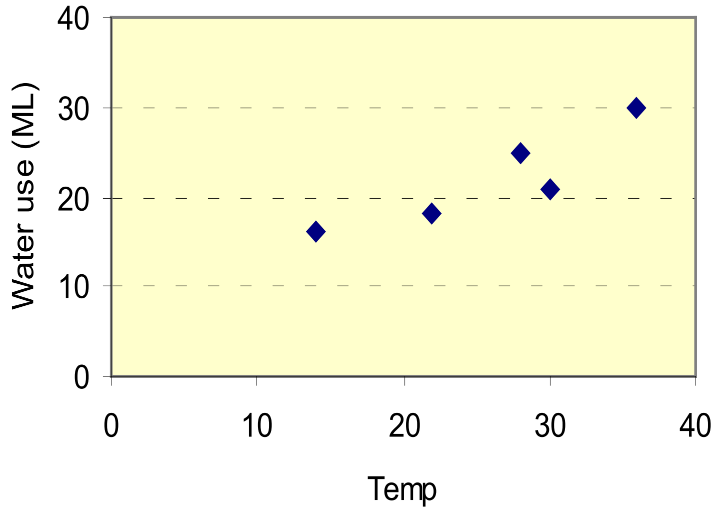
Excel

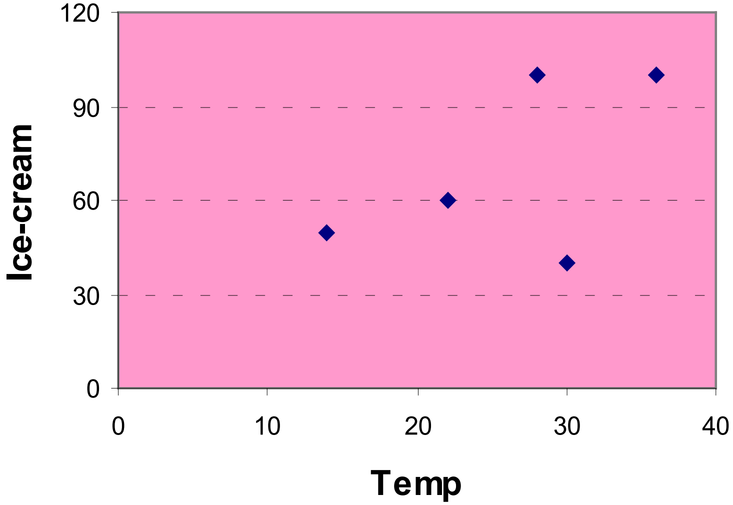
Temp	Soft Drinks
14	60
22	40
28	100
30	100
36	75

$$\text{Correlation}(X,Y) = r_{xy} = \frac{Cov_{xy}}{S_x S_y}$$ = **0.54** **=CORREL(A2:A6,B2:B6)**

Correlation Applied

Compare the relationship between temperature and beverage alternatives:





1. Temp and water use:

$$Cov(X,Y) = 42.5$$

$$s_x = 8.367 \text{ i.e. standard deviation of the x's}$$

$$s_y = 5.612 \text{ i.e. standard deviation of the y's}$$

r = Cov / (s_x*s_y) = 0.905

2. Temp and soft drink sales:

$$Cov(X,Y) = 117.5$$

$$s_x = 8.367$$

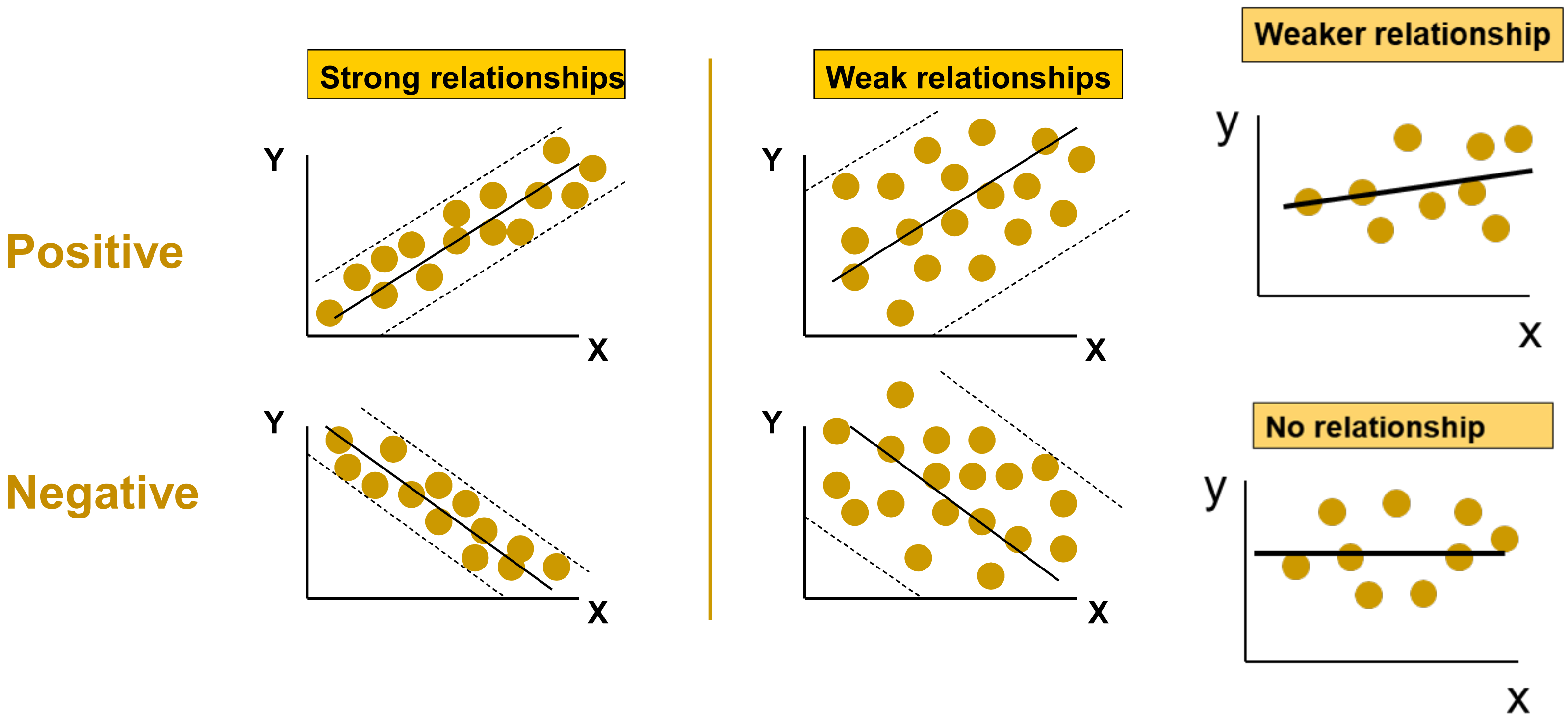
$$s_y = 25.981$$

r = Cov / (sx*sy) = 0.541

How would we interpret this result

Relationship Statistic: Correlation

At session start we considered a relationship using a scatterplot; a relationship between a dependent variable (Y) and an independent variable (X)



Task: A relationship is NOT considered to be causal; WHY?

End of Session